NAME: ________________________ CLASS: __________ INDEX: ___________

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

13 September 2021

2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing Papers and Appendix A

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question paper.

Section B
Answer **one** question in this section on writing papers provided.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten your answers for **Section B** securely.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1 [30]	
2 [9]	
3 [11]	
4 or 5 [25]	
TOTAL	75

Section A

Answer **all** questions in this section.

1 Pfizer-BioNTech mRNA COVID-19 vaccine (also known as comirnaty) is one of the first COVID-19 vaccines to receive validation from World Health Organisation (WHO) for emergency use.

Comirnaty is composed of a messenger RNA (mRNA) which is encapsulated in lipid nanoparticles (LNPs); mRNA-LNPs.

The mRNA encodes the spike (**S**) glycoprotein of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) which is a virus responsible for COVID-19.

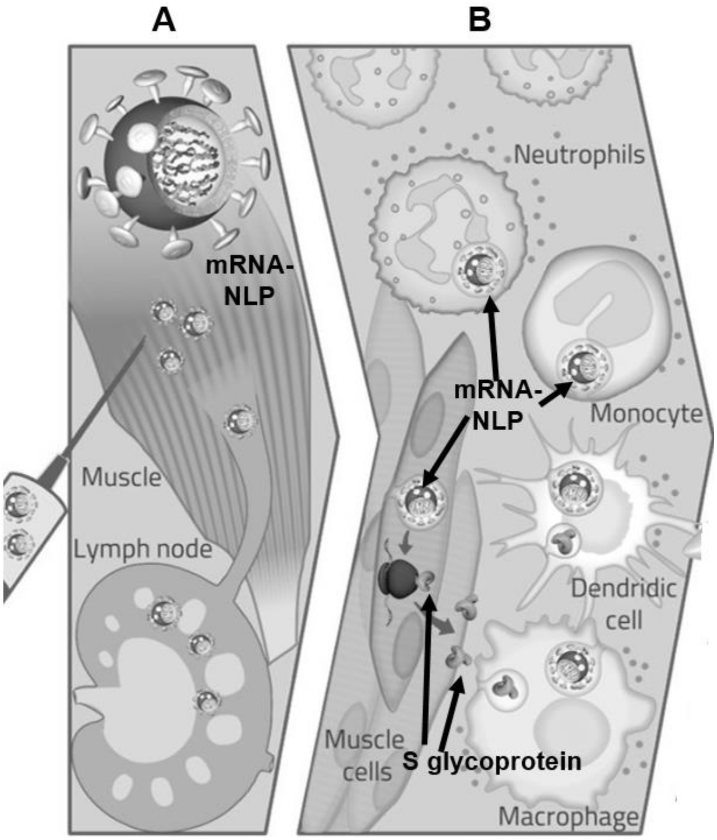

Fig.1.1 A: illustrates the administration of comirnaty vaccine intramuscularly, **B:** illustrates the uptake of mRNA-NLPs by various cells after the administration of the vaccine.

<u>*Adapted and modified from:*</u>
Armbruster N, Jasny E, Petsch B. Advances in RNA vaccines for preventive indications: A case study of a vaccine against rabies. Vaccines. 2019; 7(4):132. PubMed link

(a) Explain why the mRNA in comirnaty is encapsulated in lipid nanoparticles (LNPs).

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Fig 1.2 shows some organelles in a muscle cell.

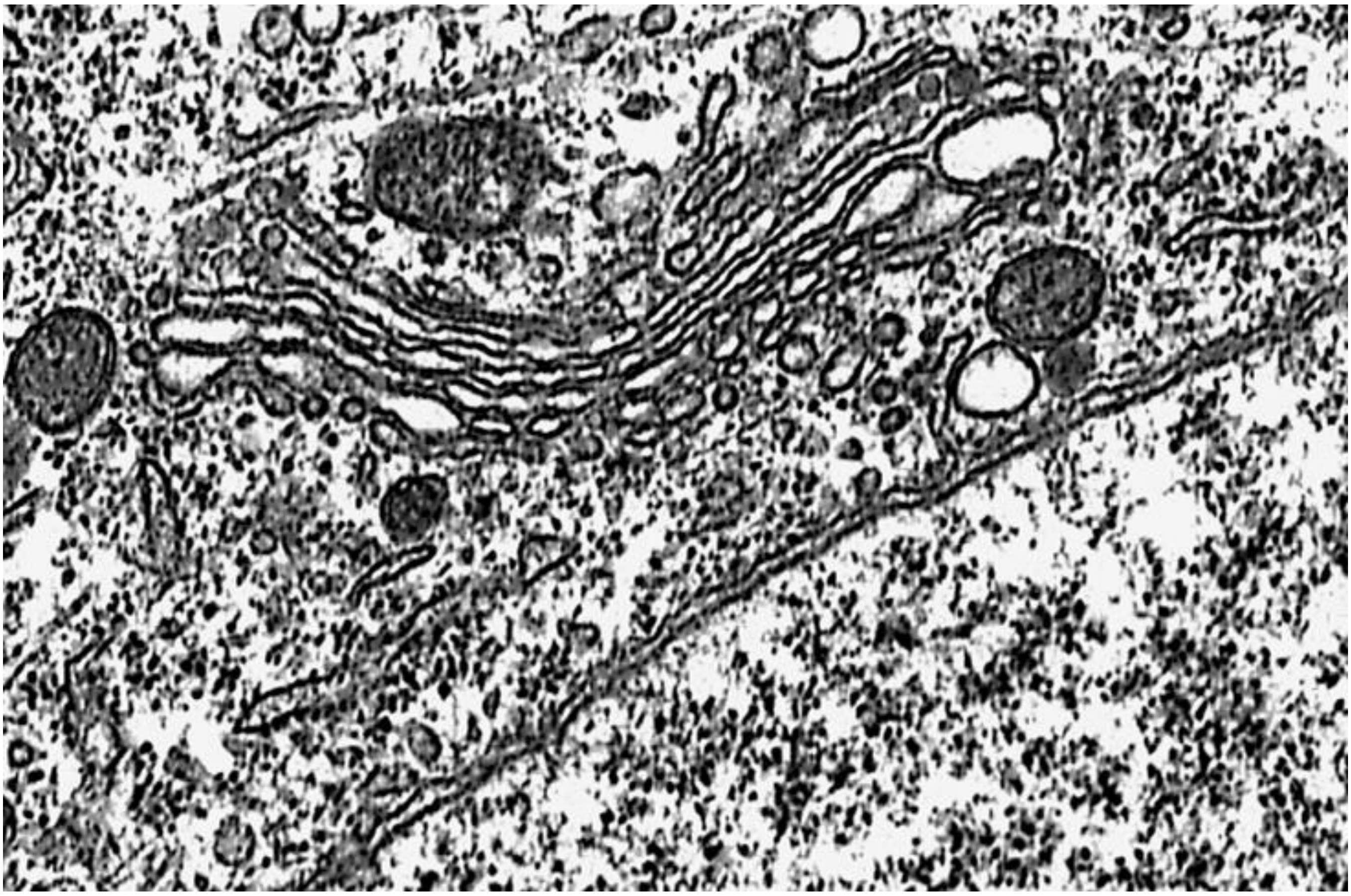

Fig. 1.2

(b) With reference to Fig. 1.2, name **two** organelles other than mitochondria that are involved the synthesis and secretion of the SARS-CoV-2 **S** glycoprotein from the muscle cell after the uptake of comirnaty into the cell. Justify your answer.

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(c) Explain how the termination phase of translation allows the newly synthesised polypeptide to be released from the translation assembly.

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(d) Explain how mitochondria is involved in the synthesis of the polypeptide of **S** glycoprotein in the muscle cells that took in the mRNA-LNPs.

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A study was performed to investigate the effect of different concentrations of oxygen (O_2) on the synthesis of **S** glycoprotein in muscle cells that took in the mRNA-LNPs. All other conditions are kept constant.

Under normal condition, the concentration of O_2 in the air is around 20%. In addition, the normal concentration of lactic acid in muscle cells is around 0.50 – 2.00 mM. Beyond 4.00 mM of lactic acid, cell death occurs.

Table 1.1 shows the result obtained for the study.

Table 1.1

Concentration of O_2 / %	**Average amount of S glycoprotein obtained from 5.0 x 10^6 muscle cells after 24 hours / μg**	**Average concentration of lactic acid obtained from 5.0 x 10^6 muscle cells after 24 hours / mM**
20	0.410	0.50
15	0.382	1.80
10	0.251	3.62
5	0.093	4.50
0	0.021	4.50

(e) With reference to Table 1.1,

(i) account for the effect of the decrease in concentration of O_2 from 20% to 15% on the synthesis of **S** glycoprotein in the muscle cells.

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(ii) suggest why muscle cells subjected to 5% and 0% of O_2 were still able to produced small amount of **S** glycoprotein despite the high concentration of lactic acid that is beyond 4.00 mM.

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Fig 1.3 shows the uptake of mRNA-LNPs into cell.

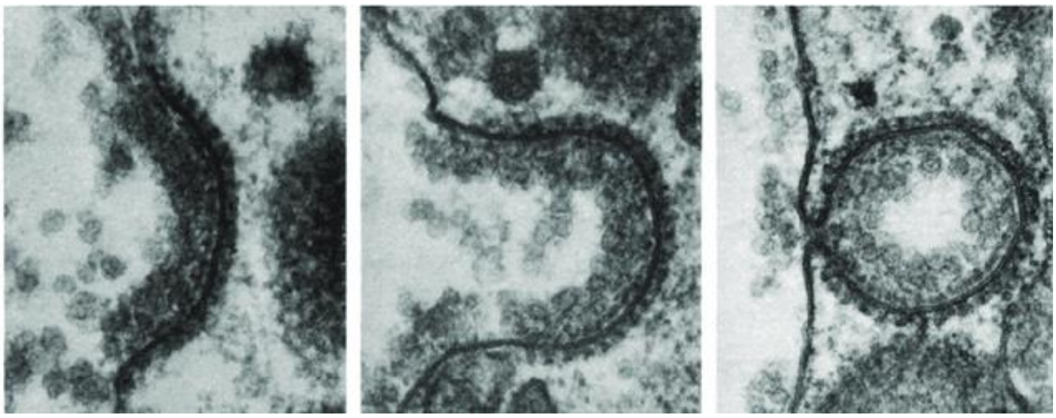

Fig. 1.3

(f) A student claimed that mRNA-LNPs do not enter cells through fusion with cell surface membrane. Instead, mRNA-LNPs enter cells through endocytosis. Based on the information in Fig. 1.3; justify the claim.

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(g) The structure of mRNA used in comirnaty vaccine resembles the structure of mRNA found in the cytoplasm of eukaryotic cells.

Explain how the structure of mRNA in comirnaty vaccine allows it to serve its role inside the muscle cells.

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The effectiveness of comirnaty vaccine was studied in two groups of population residing in region **A** and region **B** respectively. All other factors were kept constant between the two groups except for the region where they are residing.

Blood sample was taken from 5 individuals from each regions to measure the concentration of antibody against **S** glycoprotein of SARS-CoV-2.

Table 2.2 below shows the data collected from the study.

Table 2.2

Place of residence	**Concentration of antibody against S glycoprotein of SARS-CoV-2 / mM / 100 mL of blood**
Region **A**	5.2
	6.7
	5.8
	6.9
	5.5
Region **B**	6.2
	5.4
	4.8
	4.3
	6.3

(h) Using an appropriate statistical test, determine if the difference in the effectiveness of comirnaty in people residing in region **A** and region **B** is statistically significant.

Show your working in the space below.

(*You are provided with **Appendix A** which contains the required statistical formulae, t-test Distribution Table and X^2-test Distribution Table to solve this question)*

Null hypothesis : ……………………………………………………………………………

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Alternative hypothesis : ……………………………………………………………………………

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[Total: 30]

2 Genetically identical bacteria cells; *Escherichia coli* (*E.coli*) were divided equally into two groups; namely:

Control Group : the *E. coli* cells continue to be grown under optimum conditions.

Treatment Group : the *E. coli* cells continue to be grown under the same optimum conditions as in the control group. In addition, this group of *E. coli* cells are incubated with λ bacteriophages carrying ampicillin (a type of antibiotic) resistance gene (*AmpR*).

After 24 hours of incubation, the *E. coli* cells from the **Treatment Group** was found to have acquired resistance to ampicillin while the cells from the **Control Group** was lacking the resistance to ampicillin.

(a) With reference to the information above, name the horizontal gene transfer that occurred in the **Treatment Group**. Explain your answer.

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(b) Without the use of molecular techniques, suggest how the presence of *AmpR* gene in the **Treatment Group** could be inferred.

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A separate experiment was performed on the **Control Group** and **Treatment Group** mentioned above after the 24 hours of incubation.

This experiment aims to investigate the effect of the horizontal gene transfer process on the amount of chromosomal DNA in the *E. coli*.

Fig. 2.1 shows the result of the experiment.

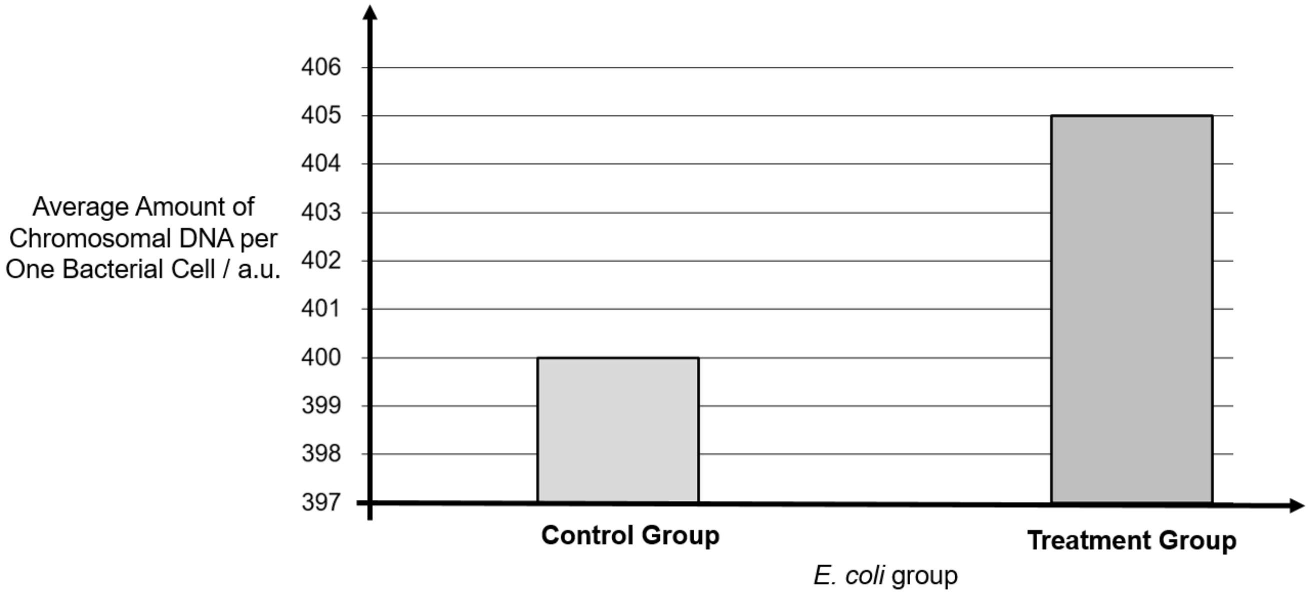

Fig. 2.1

Table 2.2 provides information on the size of various DNA.

Table 2. 2

DNA	Size in kb
E.coli chromosome	4,000
λ phage genome	48.8
AmpR gene	1.2

(c) With reference to Fig. 2.1 and Table 2.2, account for the change in the average amount of chromosomal DNA per one bacterial cell in the **Treatment Group**.

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(d) The scientists would like to amplify the *AmpR* gene using Polymerase Chain Reaction (PCR). However, the DNA sequence of *AmpR* gene is unavailable.

Suggest why PCR could not be used to amplify the *AmpR* gene.

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[Total: 9]

3 Dengue fever is caused by the dengue virus (DENV), of which there are 4 serotypes DENV1 – DENV4.

The disease is transmitted when an infected *Aedes aegypti* mosquito bites a human, injecting the virus. *Aedes aegypti* has rapidly expanded its range in the last 50 years.

Where once it was confined to the tropics, this mosquito has been found at high altitudes and higher latitudes. Global warming is now driving this mosquito out of its normal range and into new, uncharted latitudes. Along with the mosquito, DENV is now cropping up in countries that never recorded a single dengue fever case.

(a) Explain how global warming has caused the spread of DENV beyond the tropics.

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Public health efforts to counter the rise in dengue transmission during the wet season is to frequently clean out rain gutters and empty out containers and pots.

(b) Describe the conditions Aedes aegypti requires for propogation.

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In tropical countries where *Aedes aegypti's* population is already well established, small increases in overall temperature can benefit the mosquito's physiology and behaviour.

(c) Explain how climate change affects insects' physiology.

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With countries where more than one serotype of DENV is circulating, citizens are vulnerable to infections from all the 4 serotypes (DENV1 – DENV4) equally, regardless if they had been infected by different serotype before. It seems that being infected by one serotype does not give immunity against other serotypes

(d) Explain why an individuals are not immune to other serotypes after being infected with one serotype.

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(e) Describe briefly the immune response that is responsible for the fever symptom that sets in almost immediately once a patient gets infected with dengue.

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[Total: 11]

Section B

Answer **one** question in this section
.
Write your answers on separate answer paper provided.
Answer each part on a **separate** piece of paper.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

4 **(a)** Photosynthesis is a complex process**,** with reference to the structure and function of chloroplast, explain how the efficiency of photosynthesis is maximized in plants. [13]

(b) There are many factors that contribute to evolution in eukaryotes. Explain the roles of chance and isolation in an evolution that leads to the emergence of a new species. [12]

[Total: 25]

5 **(a)** "Respiration is a series of oxidation and reduction processes."

Discuss the purpose of oxidation and reduction during respiration. [13]

(b) In the human population, different combination of phenotypes are observed in each individual, except identical twins.

Explain how the different combination of phenotypes in each individual could have arisen. [12]

[Total: 25]

END OF PAPER

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